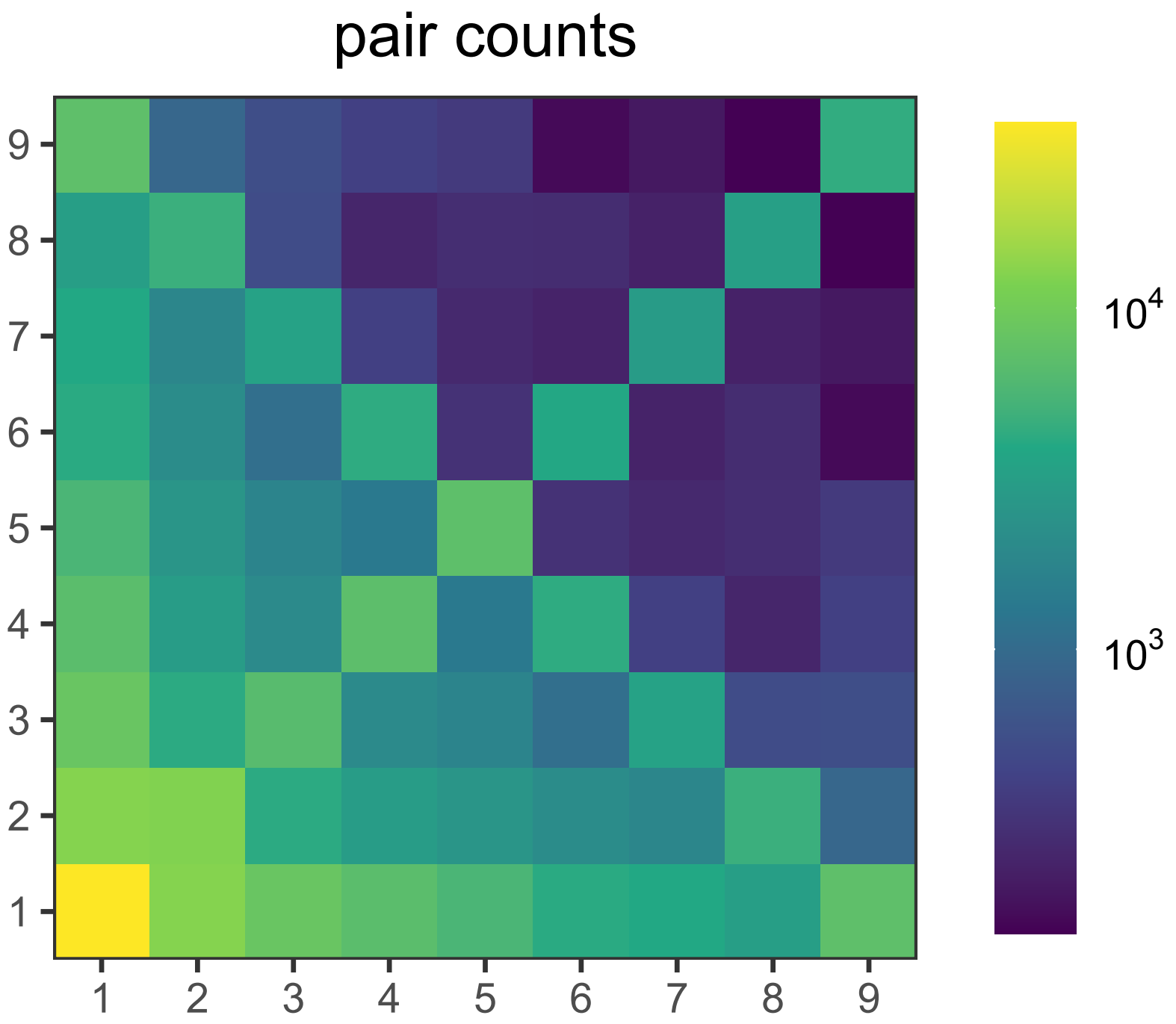
pair counts
9
8
7
6
5
4
3
2
1
1
2
3
4
5
6
7
8
9
10^4
10^3